Hypogonadotropic Hypogonadism 23 With or Without Anosmia (HH23) — Comprehensive Disease Report

Disease: Hypogonadotropic Hypogonadism 23 With or Without Anosmia **MONDO ID:** MONDO:0009223 · **OMIM:** #228300 · **Category:** Mendelian (autosomal recessive) **Causal gene:** *LHB* (luteinizing hormone subunit beta, 19q13.33)

Summary

Hypogonadotropic Hypogonadism 23 With or Without Anosmia (HH23) is, despite its name, an **autosomal-recessive isolated luteinizing-hormone (LH) deficiency** caused by biallelic loss-of-function or bioinactivating mutations in the *LHB* gene encoding the β-subunit of luteinizing hormone. The disease corresponds historically to **"Pasqualini syndrome"** and the **"fertile eunuch"** phenotype. The core lesion is the absence of a functional bioactive LH heterodimer, which leaves the intact LH/choriogonadotropin receptor (LHCGR) unstimulated and silences the downstream cAMP/PKA/StAR steroidogenic program in gonadal (Leydig and theca/granulosa) cells. GnRH secretion, FSH, the LH receptor, and the gonads themselves are intrinsically normal — the defect is confined to the LH ligand.

A critical clarification arising from this investigation: **anosmia is NOT an intrinsic feature of HH23**. The "With or Without Anosmia" phrasing is a naming artifact inherited from the OMIM/MONDO hypogonadotropic-hypogonadism numbered series (which groups Kallmann-syndrome loci alongside isolated gonadotropin-subunit defects). HH23/LHB deficiency is a **normosmic**, non-Kallmann condition. The olfactory system is unaffected because the pathology lies in gonadotrope hormone production, not in embryonic GnRH-neuron/olfactory-placode migration.

The phenotype is **sex-dimorphic and highly treatable**. Males present with severe disease — absent or incomplete puberty, delayed virilization, small testes, Leydig-cell hypoplasia, and azoospermia or markedly impaired spermatogenesis, with undetectable/low LH but normal FSH and low testosterone. Females show a milder picture — normal female external genitalia,

spontaneous breast and pubic-hair development, normal or late menarche, followed by secondary oligo-/amenorrhea and anovulatory infertility. Because the receptor and gonad are intact, **hCG (which binds the same LHCGR) restores steroidogenesis, spermatogenesis, ovulation and fertility**, and in females pulsatile GnRH or LH/hCG ovulation induction has produced healthy term pregnancies. The biochemical hallmark — **low LH with normal FSH and preserved hCG responsiveness** — distinguishes HH23 both from classic GnRH-deficient CHH (where both gonadotropins are low) and from LHCGR inactivation (where LH is high and the patient is hormone-resistant).

Key Identifiers and Disease Information

Resource	Identifier
MONDO	MONDO:0009223
OMIM	#228300 (Hypogonadotropic hypogonadism 23 with or without anosmia; HH23)
Orphanet	ORPHA:325448
DOID	DOID:0090091
MeSH	C537919
UMLS	C0271582
ICD-9	253.4
Causal gene	*LHB* (HGNC; luteinizing hormone subunit beta), chromosome 19q13.33

Synonyms / alternative names: LH deficiency; isolated luteinizing hormone deficiency; luteinizing hormone beta-subunit deficiency; **Pasqualini syndrome**; **fertile eunuch syndrome** (historical, describing the hypomorphic/partial variant with preserved spermatogenesis despite poor virilization); hypogonadism due to LH deficiency.

Information provenance: Disease-level knowledge for HH23 derives from **aggregated resources** (OMIM, Orphanet, MONDO) combined with a small set of **individual-patient case reports and small family studies** (fewer than ~20 unrelated families described worldwide), because inactivating *LHB* mutations are exceptionally rare. Ontology mapping was verified via EBI OLS; the landmark PMID was verified via NCBI E-utilities.

Key Findings

Finding 1 — HH23 is caused by biallelic inactivating *LHB* mutations (isolated LH deficiency, not Kallmann syndrome)

HH23 (MONDO:0009223 / OMIM #228300) results from **biallelic (homozygous or compound-heterozygous) loss-of-function or bioinactivating variants in *LHB***, the gene encoding the hormone-specific β-subunit of luteinizing hormone on 19q13.33. This is an **isolated LH deficiency**, mechanistically and clinically distinct from Kallmann syndrome (which is caused by genes governing GnRH-neuron/olfactory migration such as *ANOS1/KAL1*, *FGFR1*, *PROKR2*, *PROK2*, *FGF8*, *SOX10*). The "with or without anosmia" descriptor is a series-naming artifact; olfaction is intrinsically preserved.

Multiple independent human families establish the genotype spectrum:

Variant (protein / cDNA)	Zygosity	Sex / phenotype	PMID
Lys40del (in-frame deletion, trafficking determinant)	homozygous	male, severe	27656125
Trp28X (nonsense)	homozygous	male hypogonadism	29476300
IVS2+1G>C (splice)	homozygous	female, selective LH deficiency	34535998
F84L + G122S (missense)	compound het	female CHH	39786527
Hypomorphic missense (residual activity)	—	male, normal spermatogenesis	19890128
Gln54Arg (missense; first described)	homozygous	male hypogonadism	1727547
Trp8Arg + Ile15Thr	—	fertile eunuch	14709845

> *"Inactivating Luteinizing hormone beta (LHB) gene mutations are exceptionally rare and lead to hypogonadism that is particularly severe in males."* — PMID: 27656125

> *"Inactivating mutations of the luteinizing hormone beta subunit (LHB) gene are extremely rare and can result in congenital hypogonadotropic hypogonadism (CHH)." — PMID: 39786527*

Variant classification: The reported variants span nonsense (Trp28X), splice-site (IVS2+1G>C), in-frame deletion (Lys40del), and missense (Gln54Arg, F84L, G122S, Trp8Arg, Ile15Thr) classes, and are classified **pathogenic/likely pathogenic** under ACMG/AMP criteria on the basis of segregation, functional data, and absence/rarity in population databases (gnomAD). All are **germline**. Functional consequence is **loss of function** of bioactive LH — via either absent synthesis (nonsense/splice), impaired secretion/trafficking (deletion/missense), or bioinactivity of a secreted but non-signaling hormone (Gln54Arg).

Finding 2 — hCG (and pulsatile GnRH/LH in females) restores gonadal function and fertility

Because HH23 is a **ligand-replaceable deficiency with an intact receptor and gonad**, treatment is LH replacement. hCG binds and activates the same LHCGR, driving steroidogenesis and gametogenesis.

- **Male, Trp28X:** after 40 days of hCG, a few spermatozoa appeared; by 6 months, normal seminal parameters (PMID: 29476300).
- **Male brothers, Lys40del:** hCG induced pubertal development, rising testosterone, and spermatogenesis (PMID: 27656125).
- **Female, IVS2+1G>C:** ovulation induction with recombinant LH then urinary hCG achieved follicle growth, ovulation, and a healthy term pregnancy (PMID: 34535998).
- **Female, compound het F84L+G122S:** pulsatile GnRH promoted puberty and ovulation (PMID: 39786527).

> *"After 40 days of treatment with hCG, the patient exhibited a few spermatozoa in the semen. Treated for 6 months, the patient exhibited normal seminal parameters." — PMID: 29476300*
>
> *"Ovarian follicle growth and ovulation were achieved, and a healthy baby was born after an uneventful term pregnancy." — PMID: 34535998*

NCIT clinical-intervention term suggestions: chorionic gonadotropin / hCG therapy (NCIT:C617), gonadotropin-releasing hormone agonist/pulsatile GnRH, testosterone replacement therapy (for virilization when fertility is not the goal), recombinant LH.

Finding 3 — Sex-dimorphic phenotype: severe in males, milder in females

Males (severe): pubertal delay/absent virilization, small testes, azoospermia or variable spermatogenesis, undetectable/low LH with normal FSH, low testosterone, and absent or few fetal Leydig cells (**Leydig-cell hypoplasia**) ([PMIDs: 27656125, 19890128, 29476300]).

Females (milder): normal female external genitalia; spontaneous thelarche and pubarche; normal or late menarche followed by secondary **oligo-/amenorrhea and anovulatory infertility**; estradiol and progesterone reach follicular but not ovulatory/luteal levels; low LH with normal/slightly raised FSH; ovaries normal or enlarged with cysts ([PMIDs: 19129711, 17761593]).

A **variant "fertile eunuch" (Pasqualini) presentation** retains partial spermatogenesis with hypomorphic alleles (e.g., Trp8Arg+Ile15Thr; PMID: 14709845).

> *"female external genitalia, spontaneous breast and pubic hair development at puberty, and normal or late menarche followed by oligo-amenorrhea and infertility"* — PMID: 19129711
>
> *"the female phenotype of this LHB mutation is characterized by normal pubertal development, secondary amenorrhea, and infertility"* — PMID: 17761593
>
> *"Men with mutations in LHB, the gene encoding the beta subunit of luteinizing hormone (LHB), have azoospermia with absent or few fetal Leydig cells."* — PMID: 19890128

The sex dimorphism is explained developmentally: in females, fetal ovarian development and early puberty are largely FSH- and placental-hCG-dependent, so early pubertal milestones proceed; LH is required specifically for the ovulatory surge and corpus-luteum/progesterone function, so the deficit manifests as anovulation after menarche. In males, fetal Leydig-cell testosterone production (driving virilization of the external genitalia and later pubertal maturation) depends on LH/hCG, so loss causes more profound and earlier manifestations.

Suggested HPO terms: Hypogonadotropic hypogonadism (HP:0000044); Decreased circulating luteinizing hormone (HP:0011969); Delayed puberty (HP:0000823); Secondary amenorrhea (HP:0000869); Oligomenorrhea (HP:0000876); Azoospermia (HP:0000027); Male infertility (HP:0003251); Female infertility (HP:0008222); Decreased testicular size (HP:0008734); Decreased serum testosterone level (HP:0040171); Leydig cell hypoplasia (reduced testicular interstitial cells).

Finding 4 — Mechanism: LHB loss-of-function abrogates bioactive LH, silencing the LHCGR–cAMP/PKA–StAR steroidogenic cascade

LH is an **αβ heterodimer**: a shared α-subunit (encoded by *CGA*) non-covalently associated with the hormone-specific LHB β-subunit. Different *LHB* mutation classes disrupt the hormone in distinct ways:

- **Missense/in-frame deletion (e.g., Lys40del):** the mutant subunit is **retained intracellularly** (diffuse cytoplasmic distribution) and is not secreted, even though it can still heterodimerize with the α-subunit and, in vitro, activate the receptor. Lys40 lies in a conserved **trafficking determinant** shared across the hCG/FSH/TSH β-subunits (PMID: 27656125).
- **Splice/nonsense (IVS2+1G>C, Trp28X):** abolish LH synthesis or secretion outright ([PMIDs: 17761593, 29476300]).
- **Bioinactive missense (Gln54Arg):** a hormone that is immunoreactive but biologically inactive — it cannot productively engage/activate LHCGR (PMID: 1727547).

Downstream, LH/hCG act through **LHCGR**, a Gs-coupled GPCR, activating **cAMP → PKA → CREB**, which induces **StAR** (steroidogenic acute regulatory protein) and steroidogenic enzymes (*CYP11A1*, *HSD3B*) to drive testosterone/progesterone synthesis ([PMIDs: 33112765, 38621656]). Loss of bioactive LH therefore silences this cascade at its most upstream (ligand) node.

> *"the mutant LH is retained intracellularly and showed diffuse cytoplasmic distribution. The mutated LHB heterodimerizes with the common alpha-subunit and can activate its receptor."* — PMID: 27656125

Mouse models recapitulate the disease: *Lhb*-knockout mice show hypogonadism with reduced androgen-dependent testicular gene programs (e.g., *Slc7a11*), and female *Lhb*-KO mice are close phenocopies of human *LHB* mutations ([PMIDs: 35398053, 19129711]).

> *"we performed large-scale gene expression analyses on testes obtained from adult control and Lhb knockout mice"* — PMID: 35398053

CHEBI chemical entities: luteinizing hormone; human chorionic gonadotropin; testosterone (CHEBI:17347); progesterone (CHEBI:17026); estradiol (CHEBI:23965); cyclic AMP / 3',5'-cyclic AMP (CHEBI:17489); cholesterol (CHEBI:16113, StAR substrate).

GO/CL term suggestions: biological process — hormone-mediated signaling pathway (GO:0009755), steroid biosynthetic process (GO:0006694), response to luteinizing hormone, cAMP-mediated signaling (GO:0019933), male gonad development (GO:0008584), ovulation (GO:0030728); cellular component — extracellular space (secreted LH), endoplasmic reticulum (trafficking-retained mutants, GO:0005783); cell types — Leydig cell (CL:0000178), granulosa cell (CL:0000501), theca cell (CL:0000503), pituitary gonadotrope (CL:0000173).

Finding 5 — Key diagnostic discriminator: low LH + hCG-responsive (*LHB*) vs. high LH + hormone-resistant (*LHCGR*)

Women with inactivating *LHB* vs. *LHCGR* mutations share a near-identical clinical picture (female external genitalia, spontaneous breast/pubic-hair development, normal/late menarche, then oligo-amenorrhea and infertility), but are distinguished **biochemically and therapeutically**:

Feature	*LHB* deficiency (HH23)	*LHCGR* inactivation
Serum LH	**Low / undetectable**	**High**
Serum FSH	Normal / slightly raised	Normal / slightly raised
Defect location	Ligand (hormone)	Receptor
hCG/LH response	**Responsive**	**Resistant**
Treatment	hCG / LH replacement	Not receptor-correctable

> *"serum LH levels are low in patients with LHB mutations and high in those with LHCGR mutations, whereas follicle-stimulating hormone levels are normal or only slightly increased"* — PMID: 19129711
>
> *"Women with LHB mutations may be treated with hCG (human chorionic gonadotropin) or LH, whereas those with mutations in LHCGR are resistant."* — PMID: 19129711

Diagnostic pitfall — bioinactive LH: The Gln54Arg hormone is **immunoreactive but biologically inactive**. A standard immunoassay may therefore report "detectable/normal" LH that is in fact non-functional. In a patient with hypogonadism and a normal-appearing LH immunoassay, discordance between immunoreactivity and bioactivity (or molecular testing) is required to avoid missing the diagnosis (PMID: 1727547).

Finding 6 — Minimal residual intratesticular testosterone suffices for complete spermatogenesis (molecular basis of the "fertile eunuch")

A man with a **hypomorphic *LHB* mutation** had undetectable serum LH and low serum testosterone, yet **complete, quantitatively normal spermatogenesis**. The mutant LH retained low partial in-vitro activity, sufficient to drive steroidogenic-enzyme expression in a few mature Leydig cells that produced only **~20.2 ng/g intratesticular testosterone** — enough for normal spermatogenesis despite absent virilization (PMID: 19890128). This dissociation of poor virilization from preserved fertility is the molecular basis of the historical **"fertile eunuch"/ Pasqualini** phenotype (e.g., Trp8Arg+Ile15Thr; PMID: 14709845) and challenges the dogma about the high intratesticular-testosterone threshold required for spermatogenesis (PMID: 34884539).

> *"the residual luteinizing hormone activity, resulting in the expression of steroidogenic enzymes in few mature Leydig cells producing small amounts of intratesticular testosterone (20.2 ng per gram), was sufficient for complete and quantitatively normal spermatogenesis"* — PMID: 19890128
>
> *"Normal virilization and improved semen parameters were achieved after hCG supplementation."* — PMID: 14709845

Finding 7 — Historical landmark: Weiss 1992 established LH-beta deficiency as a distinct cause of hypogonadism

The first human *LHB* mutation was reported by **Weiss et al., N Engl J Med 1992;326(3):179–183** (*"Hypogonadism caused by a single amino acid substitution in the beta subunit of luteinizing hormone"*; PMID: 1727547). They described a homozygous **Gln54Arg** missense variant in a man with delayed puberty/hypogonadism. The mutant LH was immunoreactive but biologically inactive — the first molecular proof that an **intrinsic LH-beta defect** (rather than GnRH deficiency or pituitary failure) can cause hypogonadism. Bibliographic details were verified via NCBI E-utilities.

Mechanistic Model — Causal Chain

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1. Biallelic inactivating / bioinactivating LHB mutation (19q13.33)
        │  (nonsense/splice → no synthesis;  missense/deletion → ER retention / no secretion;
        │   Gln54Arg → secreted but bioinactive)
        ▼
2. Absence of functional bioactive LH αβ heterodimer in circulation
        │  leads to
        ▼
3. LHCGR (Gs-coupled GPCR) on Leydig / theca cells remains unstimulated
        │  results in
        ▼
4. Failure of cAMP → PKA → CREB signaling   [inferred from receptor/pathway biology]
        │  leads to
        ▼
5. Loss of StAR induction + reduced CYP11A1 / HSD3B activity
        │  results in
        ▼
6. Deficient gonadal steroidogenesis (low testosterone / low progesterone)
        │
        ├──► MALE branch: fetal + pubertal Leydig-cell testosterone absent
        │        → Leydig-cell hypoplasia, undervirilization, absent/incomplete
        │          puberty, azoospermia/impaired spermatogenesis
        │        (hypomorphic alleles → trace intratesticular T → "fertile eunuch")
        │
        └──► FEMALE branch: early puberty (FSH/hCG-driven) proceeds normally,
                 but absent LH surge → anovulation, no corpus luteum,
                 low luteal progesterone → secondary oligo-/amenorrhea, infertility
        ▼
7. Clinical HH23.  Because LHCGR and gonad are intact, hCG/LH replacement
   re-enters the chain at step 3 and restores steroidogenesis, gametogenesis,
   and fertility.
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Upstream vs downstream: The initiating lesion (steps 1–2) is the ligand defect; steps 3–6 are the intact-but-unstimulated receptor/steroidogenic machinery (downstream, and therefore rescuable by exogenous agonist). FSH, GnRH, and the olfactory/GnRH-migration axis are entirely outside this chain — the reason HH23 is normosmic and FSH-normal.

Comparison With Related Conditions (Differential Diagnosis)

Condition	LH	FSH	Smell	Key gene(s)	hCG response
HH23 / *LHB* deficiency	**Low**	Normal	**Normal**	*LHB*	**Yes**
Kallmann syndrome	Low	Low	**Anosmia**	*ANOS1, FGFR1, PROKR2, PROK2, SOX10*	Variable
Normosmic CHH (GnRH-deficient)	Low	Low	Normal	*GNRHR, KISS1R, TAC3/TACR3*	Variable
LHCGR inactivation (Leydig hypoplasia)	**High**	Normal	Normal	*LHCGR*	**No (resistant)**
FSHB deficiency	Normal/ high	Low	Normal	*FSHB*	N/A

The single most useful discriminator between HH23 and GnRH-deficient CHH is that **FSH is preserved in HH23** (only LH is deficient), whereas GnRH deficiency lowers both gonadotropins. The discriminator from *LHCGR* disease is the **low (vs high) LH and preserved hCG responsiveness**.

Anatomical Structures, Temporal Course, and Epidemiology

Anatomy / organ level (UBERON/CL): Endocrine axis — pituitary gonadotrope (CL:0000173) as the site of the production defect; primary target organs are the **testis** (UBERON:0000473) Leydig/interstitial cells (CL:0000178) and the **ovary** (UBERON:0000992) theca (CL:0000503) and granulosa cells (CL:0000501). Subcellular compartments: endoplasmic reticulum (trafficking-retained mutants), extracellular space (secreted hormone), and mitochondrial/ER steroidogenic machinery (StAR/CYP11A1). Secondary manifestations: reduced bone mineralization and body-composition changes expected with chronic sex-steroid deficiency (general hypogonadism complications; not disease-specific data).

Temporal development: Congenital/germline genetic basis with **onset recognized at the expected pubertal window** — males present with absent/incomplete puberty and small testes; females present at or after the normal menarcheal age with secondary amenorrhea. Course is **chronic and lifelong** at the level of the endogenous deficiency but is **not progressive**; it is fully controllable with replacement therapy. The pubertal window is the **critical period** for diagnosis and for virilization/breast development; fertility can be induced later with hCG/LH or pulsatile GnRH.

Epidemiology: Inactivating *LHB* mutations are **"exceptionally rare"** (PMID: 27656125) — fewer than ~20 unrelated families reported globally; no reliable prevalence/incidence estimate exists (ultra-rare). **Inheritance is autosomal recessive**; most reported probands are homozygous, several from consanguineous unions, implying a contributing role for consanguinity and founder/private alleles. **Penetrance** of biochemical hypogonadism appears high in biallelic carriers, with **variable expressivity** (severe azoospermia through fertile-eunuch depending on residual allele activity). Heterozygous carriers are clinically unaffected. No genetic anticipation (not a repeat-expansion disorder). Both sexes affected; males are more severely affected.

Diagnostics (Synthesis)

- **Hormonal profile:** low/undetectable LH, **normal FSH**, low sex steroids (testosterone in males; failure to reach ovulatory estradiol/luteal progesterone in females). This FSH-normal, LH-low pattern is the biochemical signature.
- **Caution:** some bioinactive mutants (Gln54Arg) are **immunoreactive**, so immunoassay LH may appear normal — bioassay discordance or molecular testing resolves this (PMID: 1727547).
- **hCG stimulation test:** a rise in testosterone/steroids confirms an intact receptor/gonad and supports *LHB* (vs *LHCGR*) etiology (PMID: 19129711).
- **Genetic testing:** single-gene *LHB* sequencing, or CHH/hypogonadism gene panels / whole-exome sequencing. Given ultra-rarity and phenotype overlap, panel/WES is practical; confirm biallelic status and segregation.
- **Imaging:** small testes on exam/ultrasound (males); normal or cystic/enlarged ovaries (females). Normal olfaction on formal smell testing distinguishes from Kallmann syndrome; normal MRI of olfactory bulbs expected (not routinely required).

Treatment, Prognosis, and Prevention

Treatment. HH23 is among the most treatable hypogonadotropic disorders because the receptor and gonad are intact: - **Virilization / sex-steroid replacement:** testosterone (males), estrogen/progestin (females) for secondary sexual characteristics and bone health when fertility is not the immediate goal. - **Fertility induction: hCG** (acts on LHCGR) ± FSH in males to initiate/restore spermatogenesis ([PMIDs: 29476300, 27656125, 14709845]); **recombinant LH / hCG ovulation induction** or **pulsatile GnRH** in females ([PMIDs: 34535998, 39786527]). - Documented outcomes include normalization of seminal parameters within months and a healthy term pregnancy — **excellent prognosis** for both sexual maturation and fertility with treatment.

Prognosis. Life expectancy is normal; morbidity relates to untreated hypogonadism (infertility, reduced bone density, psychosocial impact of delayed puberty) and is largely reversible with therapy. No disease-specific mortality.

Prevention / counseling. No primary prevention (germline Mendelian disorder). **Genetic counseling** for autosomal-recessive recurrence risk (25% in carrier × carrier couples), **cascade carrier testing** of relatives, and reproductive options (prenatal/preimplantation genetic testing) are appropriate given identified familial variants. Carrier frequency is presumably very low (ultra-rare alleles; check gnomAD for specific variants).

Model Organisms

Model	Type	Recapitulation	PMID
Lhb-knockout mouse	Mammalian KO	Hypogonadism; reduced androgen-dependent testicular programs (e.g., *Slc7a11*); female KO is a close phenocopy of human *LHB* disease	35398053, 19129711
LuRKO (Lhcgr-KO) mouse	Mammalian KO	Receptor-side model; elevated LH, contrasts the ligand-side *Lhb* defect	25039914
Zebrafish *lhb* mutant	Non-mammalian	Normal gonadal growth but females fail to spawn (infertile); LH role in final maturation	25396299

Orthologs: *Lhb* (mouse), *lhb* (zebrafish). **Limitations:** rodent fetal Leydig-cell and placental-hCG biology differ from human; zebrafish lack the mammalian hCG axis, so the human sex-dimorphism and fertile-eunuch phenotype are not fully reproduced. No naturally occurring companion-animal/wildlife *LHB*-deficiency disease is well documented (OMIA); no zoonotic dimension (non-infectious).

Evidence Base

PMID	Role in this report	Evidence type
1727547	First human *LHB* mutation (Gln54Arg); bioinactive-LH concept	Human case (landmark)
27656125	Lys40del; trafficking determinant; rarity; male severity; hCG response	Human + in vitro
29476300	Trp28X nonsense; hCG restores spermatogenesis	Human case
34535998	Female splice variant; ovulation induction → term pregnancy	Human case
39786527	Female compound-het; pulsatile GnRH; CHH framing	Human case
19890128	Hypomorphic allele; trace intratesticular T → normal spermatogenesis	Human + functional
14709845	Fertile-eunuch (Trp8Arg+Ile15Thr); hCG response	Human case
19129711	*LHB* vs *LHCGR* discriminator (low vs high LH; responsive vs resistant); female phenotype	Human review
17761593	Female phenotype: normal puberty → amenorrhea/ infertility	Human case
35398053	*Lhb*-KO mouse testicular gene programs	Mouse model
33112765, 38621656	LHCGR → cAMP/PKA/CREB → StAR steroidogenesis pathway	In vitro
34884539	Intratesticular-testosterone threshold dogma reconsidered	Review

Limitations and Knowledge Gaps

1. **Ultra-rarity.** The entire human evidence base is a handful of case reports/small families; no epidemiological prevalence, no natural-history cohort, and no quantitative penetrance/expressivity estimates exist. Frequency statements are qualitative.
2. **Name/ontology artifact.** The OMIM/MONDO series name ("with or without anosmia") implies an olfactory feature that is not a genuine part of *LHB* deficiency; downstream knowledge bases may mis-associate HH23 with Kallmann phenotypes. This report explicitly flags anosmia as non-intrinsic.
3. **Mechanistic inference.** The cAMP/PKA/StAR silencing step (chain step 4) is inferred from LHCGR receptor biology and model systems rather than demonstrated directly in *LHB*-deficient human gonadal tissue.
4. **Modifier genes, epigenetics, and QoL** are unstudied for this specific disorder; no disease-specific quality-of-life, modifier-allele, or methylation data are available.
5. **Genotype–phenotype granularity.** The allele-activity-to-phenotype relationship (severe vs fertile-eunuch) is supported by individual cases but lacks a systematic functional allelic series.

Proposed Follow-up Actions

1. **Curate an international *LHB*-mutation registry** to aggregate the scattered case reports and derive penetrance, expressivity, and genotype–phenotype (allele residual activity vs virilization/fertility) relationships.
2. **Standardized functional assays** (secretion/trafficking + LHCGR activation) for each reported and novel *LHB* variant to build a quantitative allelic series predicting fertile-eunuch vs severe phenotypes.
3. **Correct the ontology annotation** in downstream knowledge bases to mark anosmia as *not* a feature of MONDO:0009223, preventing propagation of the Kallmann-series naming artifact.
4. **Prospective fertility-outcome data** on hCG/LH and pulsatile-GnRH protocols specific to *LHB* deficiency, to formalize treatment algorithms.
5. **gnomAD interrogation** of each pathogenic *LHB* allele to estimate carrier frequencies and inform carrier-screening/counseling.

Report compiled from 7 confirmed findings and 39 reviewed papers across 5 investigation iterations. All quoted abstract snippets were validated against stored abstracts; ontology identifiers verified via EBI OLS and NCBI E-utilities.
